Introduction to paleoclimatology

Background reading for dendrochronology · introductory working draft

Why study past climate?

Paleoclimatology investigates past variations in temperature, the water cycle and other aspects of climate using natural and human records. It extends beyond the period of instrumental measurements and examines the size, speed and regional differences of change. Past records help test our understanding of climate and place recent changes in a longer context.

Observations and proxies

When past temperature cannot be measured directly with a thermometer, a climate-sensitive feature can serve as a proxy. Tree-ring width, isotopic ratios in ice and pollen in sediment are examples. None is itself a direct measurement of temperature or rainfall. Researchers examine non-climatic influences, preservation and measurement methods before relating a proxy to a climate variable.

Questions to ask

What was measured? When and where was the record formed? Which season does it represent? What can and cannot be inferred?

Why reconstruct past climate?

Looking beyond a short instrumental record

Instrumental temperature and rainfall measurements are invaluable, but they cover only part of climate history. Older tree rings and sediments can help examine recurring variability and prolonged droughts before measurements began. This provides a longer context for assessing recent events in a region. Archives differ in temporal resolution, so short extremes may not be captured.

Understanding climate and its impacts

Comparing seasonal and regional changes with ocean and atmospheric processes or volcanic activity can test hypotheses about climate mechanisms. Researchers can compare reconstructions with climate-model simulations to assess understanding and identify areas for improvement, while checking data dependencies and uncertainties.

Combining evidence about past moisture with records of forest growth, lake levels and water use helps explore responses of ecosystems and water resources. Comparisons with historical records must also consider age, region, institutions and other possible causes rather than attributing a social change to climate alone.

Comparing past and future

A past interval is not automatically an exact analogue for the future. Greenhouse gases, volcanic forcing and land use can differ. Check the season and region, age uncertainty, the variability a reconstruction can recover, and the independence of the evidence. Multiple lines of evidence are valuable when their differences are understood.

Comparing climate archives

Examples

Tree rings: width, density or chemical composition. Crossdating can establish calendar years, but the record depends on the growing season, species and site.

Ice cores: isotope ratios in ice and gases trapped in bubbles. Record length, temporal resolution and dating uncertainty vary with depth and location.

Lake and marine sediments: pollen, microfossils and chemical compounds. Researchers consider deposition and mixing and constrain ages using several methods.

Corals and cave deposits: chemical composition of growth layers. The climate signal and temporal resolution depend on the sample and measurement.

Combining records

Archives differ in region, season and age uncertainty. Disagreement need not mean one is wrong: the records may respond to different seasons or processes. Do not count two papers using the same underlying data as two independent records. Instead, check how each record was obtained. Papers with different data may also share assumptions or calibration sources. Independent does not automatically mean error-free.

Dating and temporal resolution

An age model

For sediment, ages measured at several depths and the sequence of layers can constrain a relationship between depth and age. Each position has an age uncertainty that can vary through the core. For tree rings, researchers match patterns among trees and connect them to a calendar-dated reference chronology.

Three different questions

Sampling interval: how often was material measured? Temporal resolution: how closely spaced changes can the record actually distinguish? Age precision and uncertainty: how well can the time of a sample be established? Even closely spaced measurements can be smoothed by mixing during deposition.

A calendar-dated ring does not record the temperature of every day of that year. Trees respond to conditions during formation and may also reflect influences from previous years.

From rings to a climate estimate

The research sequence

Define the question and season, then select species and sites. Crossdate the samples and consider how age- and size-related growth trends will be treated before building a chronology. Check the location, period, variable and units of the instrumental record used for comparison.

Calibration estimates a relationship using part of the observations. Verification evaluates it with information not used to build the model. Using verification data to choose the target season or model can make performance look too optimistic. Consider how independent split-period tests and comparisons with other locations or archives actually are.

Limits and uncertainty

Standardization can remove long-term variation, depending on the method. Tree growth and temperature relationships may also change over time or space. State the region, season, period, sample depth, method and uncertainty, and distinguish observations from reconstructions.

A study and a historical question

Spring in Northeast Asia

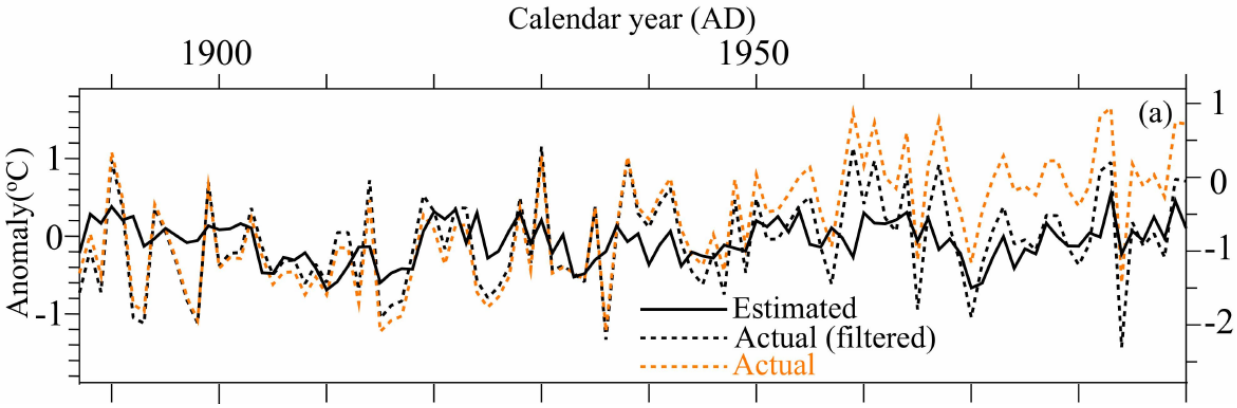
Ohyama and colleagues (2013) used five chronologies from Japan and Korea to reconstruct March–May mean land temperature for 35–40° N and 125–140° E over 1784–1990. Adjusted R^2 for calibration over 1887–1990 was 0.194. Split-period verification, swapping the earlier and later halves, gave CE values of 0.019 and −0.040. These are different statistics.

The authors treated long-term variation in the observed temperature series and used the reconstruction to study year-to-year to multidecadal variability. It should not be interpreted as a direct reconstruction of the long-term warming trend. Read the paper.

Climate and history

Overlap between a cool period and a social change does not establish causation. Check the region, season and age uncertainty of the reconstruction. Compare documentary and archaeological evidence for pathways involving food, water, institutions and land use, as well as alternative explanations.

Reading observations and estimates



Three lines and two vertical axes

The solid black line is the estimate; the dotted black line is the observation with its long-term component treated. Both use the left axis. The dotted orange line is the untreated observation and uses the right axis. The axes differ, so do not compare line heights as if they shared a scale.

The full 1887–1990 period was used for calibration. This panel is not an independent verification test. The split-period CE values discussed on the previous page are separate results.

Source: Ohyama et al. (2013), cropped from Fig. 3(a) / CC BY 3.0

Exercise and further reading

Consider this example

Samples from a lake core were measured every ten years. Mixing in the sediment has smoothed variations on a roughly 50-year scale, and the ages have uncertainty. Can these ten-year measurements identify a drought in a particular single year with certainty? This example is synthetic.

Explanation

No. Sampling interval, effective temporal resolution and age uncertainty are different. Account for mixing and the age range, then compare records with different properties, such as tree rings. Also check the growing season and non-climatic influences represented by the rings.

Explore further

NOAA: tree-ring data · NOAA: proxy data · IPCC: evidence of climate change · Ohyama et al. (2013)

Distinguish measurements, the authors' reconstruction and summaries written by others. Read the methods and limitations as well as the conclusions.